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Core Concepts

Numbers to Know

Learn about the numbers you need to know for system design interviews.

Our industry moves fast. The hardware we build systems on evolves constantly, which means even recent textbooks can become outdated quickly. A book published just a few years ago might be teaching patterns that still make sense, but quoting numbers that are off by orders of magnitude.

One of the biggest giveaways that a candidate has book knowledge but no hands-on experience during a system design interview is when they rely on outdated hardware constraints. They do scale calculations using numbers from 2015 (or even 2020!) that dramatically underestimate what modern systems can handle. You'll hear concerns about database sizes, memory limits, and storage costs that made sense then, but would lead to significantly over-engineered systems today.

This isn't the candidate's fault – they're doing the right thing by studying. But understanding modern hardware capabilities is crucial for making good system design decisions. When to shard a database, whether to cache aggressively, how to handle large objects – these choices all depend on having an accurate sense of what today's hardware can handle.

Let's look at the numbers that actually matter in 2025.

Modern Hardware Limits

Modern servers pack serious computing power. An AWS **M6i.32xlarge** comes with 512 GiB of memory and 128 vCPUs for general workloads. Memory-optimized instances go further: the **X1e.32xlarge** provides 4 TB of RAM, while the **U-24tb1.metal** reaches 24 TB of RAM. This shift matters because many applications that once required distributed systems can now run on a single machine.

Storage capacity has seen similar growth. Modern instances like AWS's **i3en.24xlarge** provide 60 TB of local SSD storage. If you need more, the **D3en.12xlarge** offers 336 TB of HDD storage for data-heavy workloads. Object storage like **S3** is effectively unlimited, handling petabyte-scale deployments as a standard practice. The days of storage being a primary constraint are largely behind us.

Network capabilities haven't stagnated either. Within a datacenter, 10 Gbps is standard, with high-performance instances supporting up to 20 Gbps. Cross-region bandwidth typically ranges from 100 Mbps to 1 Gbps. Latency remains predictable: 1-2ms within a region, and 50-150ms cross-region. This consistent performance allows for reliable distributed system design.

These aren't just incremental improvements – they represent a step change in what's possible. When textbooks talk about splitting databases at 100GB or avoiding large objects in memory, they're working from outdated constraints. The hardware running our systems today would have been unimaginable a decade ago, and these capabilities fundamentally change how we approach system design.

Applying These Numbers in System Design Interviews

Let's look at how these numbers impact specific components and the decisions we make when designing systems in an interview.

Caching

In-memory caches have grown exponentially in both size and capability. Gone are the days of 32-64GB Redis instances that required careful memory management and partial dataset caching. Today's caches routinely handle terabyte-scale datasets with single-digit millisecond latency, and a single instance can process hundreds of thousands of operations per second. This shift in scale changes the entire approach to caching strategy.

Numbers to know:

- Memory: Up to 1TB on memory-optimized instances, with some configurations exceeding this for specialized use cases
- Latency
  - Reads: < 1ms within the same region
  - Writes: 1-2ms average cross-region for optimized systems
- Throughput
  - Reads: Over 100k requests/second per instance for in-memory caches like ElastiCache Redis on modern Graviton-based nodes
  - Writes: Sustained throughput of hundreds of thousands of requests per second

When to consider sharding:

- Dataset Size: Approaching 1TB in size
- Throughput: Sustained throughput of 100k+ ops/second
- Read Latency: Requirements below 0.5ms consistently (if being exceeded, consider sharding)

These capabilities fundamentally change caching strategy. The ability to cache entire databases in memory, even at hundreds of gigabytes, means you can often avoid complex partial caching schemes altogether. This "cache everything" approach, while seemingly brute force, typically costs less than engineering time spent on selective caching logic. When you do need to scale, the bottleneck is usually operations per second or network bandwidth, not memory size – a counterintuitive shift from just a few years ago.

Databases

The raw power of modern databases surprises even experienced engineers. Single PostgreSQL or MySQL instances now routinely handle dozens of terabytes of data while maintaining millisecond-level response times. This isn't just about storage either. Modern databases efficiently handle tens of thousands of transactions per second on a single primary, with the bottleneck often being operational concerns rather than performance limits.

Numbers to know:

- Storage: Single instances handle up to 64 TiB (terabytes) for most database engines, with Aurora supporting up to 128 TiB in some configurations
- Latency
  - Reads: 1-5ms for cached data, 5-30ms for disk (optimized configurations for RDS and Aurora)
  - Writes: 5-15ms for commit latency (for single-node, high-performance setups)
- Throughput
  - Reads: Up to 50k TPS in single-node configurations on Aurora and RDS
  - Writes: 10-20k TPS in single-node configurations on Aurora and RDS
- Connections: 5-20k concurrent connections, depending on database and instance type

When to consider sharding:

- Dataset Size: Approaching or exceeding 50 TiB may require sharding or distributed solutions
- Write Throughput: Consistently exceeding 10k TPS indicates scaling considerations
- Read Latency: Requirements below 5ms for uncached data may necessitate optimization
- Geographic Distribution: Cross-region replication or distribution needs
- Backup/Recovery: Backup windows that stretch into hours or become operationally impractical

While the largest systems in the world (social networks, e-commerce giants, etc.) absolutely need sharding to handle their scale, many candidates jump to distributed solutions too early. For systems handling millions or even tens of millions of users, a well-tuned single database can often handle the load. When you do need to scale, carefully consider what's driving the decision: is it pure data volume, operational concerns like backup windows, or the need for geographic distribution? Understanding these tradeoffs leads to better scaling decisions.

> ⚠ More often then not I see candidates reaching for scaling too quickly. They have 500GB or a couple of terabytes of data and they're start explaining how they'd shard the database. Slow down, do the math, and make sure sharding is actually needed before you start explaining how you'd do it.

Application Servers

Modern application servers have evolved beyond the resource constraints that shaped many traditional design patterns. Today's servers routinely handle thousands of concurrent connections with modest resource usage, while cloud platforms enable near-instant scaling in response to load. CPU processing power, rather than memory or connection limits, typically determines your server's capabilities.

Numbers to know:

- Connections: 100k+ concurrent connections per instance for optimized configurations
- CPU: 8-64 cores
- Memory: 64-512GB standard, up to 2TB available for high-memory instances
- Network: Up to 25 Gbps bandwidth in modern server configurations
- Startup Time: 30-60 seconds for containerized apps

When to consider sharding:

- CPU Utilization: Consistently above 70-80%
- Response Latency: Exceeding SLA or critical thresholds
- Memory Usage: Trending above 70-80%
- Network Bandwidth: Approaching 20 Gbps

The implications for system design are significant. While the trend toward stateless services is valuable for scaling, don't forget that each server has substantial memory available. Local caching, in-memory computations, and session handling can all leverage this memory to improve performance dramatically. CPU is almost always your first bottleneck, not memory, so don't shy away from memory-intensive optimizations when they make sense. When you do need to scale, cloud platforms can spin up new instances in seconds, making aggressive auto-scaling a viable alternative to over-provisioning. This combination of powerful individual instances and rapid scaling means you can often achieve high performance through simple architectures.

Message Queues

Message queues have transformed from simple task delegation systems into high-performance data highways. Modern systems like Kafka process millions of messages per second with single-digit millisecond latency, while maintaining weeks or months of data. This combination of speed and durability has expanded their role far beyond traditional async processing.

Numbers to know:

- Throughput: Up to 1 million messages/second per broker in modern configurations
- Latency: 1-5ms end-to-end within a region for optimized setups
- Message Size: 1KB-10MB efficiently handled
- Storage: Up to 50TB per broker in advanced configurations
- Retention: Weeks to months of data, depending on disk capacity and configuration

When to consider sharding:

- Throughput: Nearing 800k messages/second per broker
- Partition Count: Approaching 200k per cluster
- Consumer Lag: Consistently growing, impacting real-time processing
- Cross-Region Replication: If geographic redundancy is required

The performance characteristics of modern queues challenge traditional system design assumptions. With consistent sub-5ms latencies, you can now use queues within synchronous request flows—getting the benefits of reliable delivery and decoupling without forcing APIs to be async. This speed, combined with practically unlimited storage, means queues can serve as the backbone for event sourcing, real-time analytics, and data integration patterns that previously required specialized systems.

Cheat Sheet

Here is a one-stop-shop for the numbers you need to know in 2025. These numbers represent typical values for well-tuned systems with specific workloads - your requirements may vary based on workload, hardware, and configuration. Use them as a starting point for capacity planning and system design discussions, not as hard limits. Remember that cloud providers regularly update their offerings, so while I'll try to keep this up to date, it should be treated more as a starting point than a hard limit.

Component	Key Metrics	Scale Triggers
Caching	- ~1 millisecond latency - 100k+ operations/second - Memory-bound (up to 1TB)	- Hit rate < 80% - Latency > 1ms - Memory usage > 80% - Cache churn/thrashing
Databases	- Up to 50k transactions/second - Sub-5ms read latency (cached) - 64 TiB+ storage capacity	- Write throughput > 10k TPS - Read latency > 5ms uncached - Geographic distribution needs
App Servers	- 100k+ concurrent connections - 8-64 cores @ 2-4 GHz - 64-512GB RAM standard, up to 2TB	- CPU > 70% utilization - Response latency > SLA - Connections near 100k/instance - Memory > 80%
Message Queues	- Up to 1 million msgs/sec per broker - Sub-5ms end-to-end latency - Up to 50TB storage	- Throughput near 800k msgs/sec - Partition count ~200k per cluster - Growing consumer lag

Common Mistakes In Interviews

Premature sharding

The single biggest mistake I see candidates make is assuming sharding is always necessary. They introduce a data model and immediately explain which column they'd shard on. It comes up almost every time with **Design Yelp** in particular. Here we have 10M businesses, each of which is roughly 1kb of data. This is `10M * 1kb = 10GB` of data! 10x it to account for reviews which we can store in the same database and you're only at `100GB`, why would you shard?

The same thing comes up a lot with caches. Take a **LeetCode** leaderboard where we have 100k competitions and up to 100k users per competition. We're looking at `100k * 100k * 36b ID + 4b float rating = 400GB`. While even more than what we store on disk with Yelp, this can still fit on a single large cache -- no need to shard!

Overestimating latency

I see this most with SSDs. Candidates tend to vastly overestimate the latency additional to query an SSD (Database) for a simple key or row lookup. We're talking 10ms or so. It's fast! Candidates will oftentimes justify adding a caching layer to reduce latency when the simple row lookup is already fast enough -- no need to add additional infrastructure.

> ⚠ Note, this is only for simple row lookups with an index. It is still wise to cache expensive queries.

Over-engineering given a high write throughput

Similar to the above, incorrect estimates routinely lead to over-engineering. Imagine we have a system with 5k writes per second. Candidates will often jump to adding a message queue to buffer this "high" write throughput. But they don't need to!

Let's put this in perspective. A well-tuned Postgres instance with simple writes can handle 20k+ writes per second. What actually limits write capacity are things like complex transactions spanning multiple tables, write amplification from excessive indexes, writes that trigger expensive cascading updates, or heavy concurrent reads competing with writes. If you're just inserting rows or doing simple updates with proper indexes, there's no need for complex queueing systems at 5k WPS.

Message queues become valuable when you need guaranteed delivery in case of downstream failures, event sourcing patterns, handling write spikes above 50k+ WPS, or decoupling producers from consumers. But they add complexity and should be justified by actual requirements. Before reaching for a message queue, consider simpler optimizations like batch writes, optimizing your schema and indexes, using connection pooling effectively, or using async commits for non-critical writes.

The core point is to understand your actual write patterns and requirements before adding infrastructure complexity. Modern databases are incredibly capable, and simple solutions often perform better than you might expect.

Conclusion

Modern hardware capabilities have fundamentally changed the calculus of system design. While distributed systems and horizontal scaling remain necessary for the world's largest applications, many systems can be significantly simpler than what traditional wisdom suggests.

Understanding these numbers helps you make better scaling decisions:

- Single databases can handle terabytes of data
- Caches can hold entire datasets in memory
- Message queues are fast enough for synchronous flows (as long as there is no backlog!)
- Application servers have enough memory for significant local optimization

The key insight isn't that vertical scaling is always the answer – it's knowing where the real limits are. This knowledge helps you avoid premature optimization and build simpler systems that can grow with your needs. In system design interviews, demonstrating this understanding shows that you can balance theoretical knowledge with practical experience – a crucial skill, especially for the more senior levels.

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BiologicalMoccasinTahr305 • 7 months ago
Can you give an example of where you would use message queues for synchronous flows?
5 Reply

Evan King • 7 months ago
Its still not very popular, but showing up more and more. Check this out: https://www.rabbitmq.com/tutorials/tutorial-six-python
6 Reply

Harika Reddy • 7 months ago
I currently use these RPC blocking calls in my product for interservice communication
3 Reply

Evan King • 7 months ago
Nice!
0 Reply

cst labs • 15 days ago
https://medium.com/@sinha.k/system-design-using-asynchronous-pipelines-to-handle-synchronous-flows-19c498de9cc6
0 Reply

cst labs • 15 days ago
I'd say it is not very uncommon when write throughput is way higher than server can handle but the processing may require an external service. Writing to db won't make sense in this scenario if external service needs to poll. The two can be decoupled using a RPC queue where external service can poll the queue and write response to a reply-queue.
0 Reply

vishal bajoria • 7 months ago
Thank you for this excellent article! I truly appreciate the effort in putting this together. I have a question regarding modern hardware capabilities and caching best practices. With modern RAM capacities reaching up to 24 TB, why is it recommended to limit Redis cache to 1 TB and shard when memory usage approaches 80% of this limit? Should we consider leveraging higher RAM capacities for Redis, or are there specific constraints that justify the 1 TB recommendation?
8 Reply

Evan King • 7 months ago
No strict rule here, but large instances become harder to back up, replicate, or recover from failures
16 Reply

Stefan Mai • 7 months ago
Also remember for Redis specifically that it's single-threaded! You'll probably be CPU-bound first. Other implementations can help with this (e.g. Dragonfly) but otherwise you'll need to shard anyways to increase CPU utilization.
21 Reply

vishal bajoria • 7 months ago
thank you. it makes sense.
1 Reply

BigGoldStoat775 • 7 months ago
Thank you so much, a much-needed article. Provides a lot of explicit context for the decisions that more experienced people intuitively make.
5 Reply

CuddlyCopperWildebeest704 • 7 months ago
If an application server nowadays can handle 100k+ connections, why should we consider sharding at only
Connection Count: Nearing 15k per instance?
4 Reply

Evan King • 7 months ago
Because I made an edit to the number of connections and forgot to update the scaling suggestion :) fixing
7 Reply

NeutralCrimsonCapybara540 • 6 months ago
It still says 15k connection as scaling trigger?
1 Reply

CapableEmeraldPiranha843 • 6 months ago
Then what would the correct number be?
0 Reply

PositiveEmeraldParakeet596 • 6 months ago
15K is still there
0 Reply

CuddlyCopperWildebeest704 • 7 months ago
"Modern systems like Kafka process millions of messages per second with single-digit millisecond latency, while maintaining weeks or months of data. "

This is very interesting. I was actually thinking about it in the context of top k videos where throughput is 700k/s

Assuming
1Mb/message *1M messages/s = 1TB per second
1TB/s * 30days/month * 100,000s/day = 3 000 000TB/month
That would mean Kafka for this scenario would require 3000PB * 2(replication x2 => follower and leader) = 6000 petabytes of storage?
This would require a lot of brokers, wouldn't it? Or am I missing something
1 Reply

Yaroslav Shevchuk • 6 months ago
Why would you want to post 1Mb messages?
4 Reply

CuddlyCopperWildebeest704 • 6 months ago
this is upper bound since best practice for Kafka best practice recommends max size of msg to be 1MB.
0 Reply

ZealousFuchsiaLeopon345 • 4 months ago
But your message size won't actually be 1MB (10^6 bytes), would it?
0 Reply

toby • 1 month ago
1TB per second over the network on a single machine? Good luck.
Kafka may handle 100M messages/sec at maybe 1KB per message or less, but anything beyond is into some pretty serious networking speeds. AWS currently has a c6in.32xl which can handle 200GB networking (I think by bonding cards), but that seems pretty specialised.
1 Reply

CuddlyCopperWildebeest704 • 1 month ago
good point! did not think about networking limit. thank you!
0 Reply

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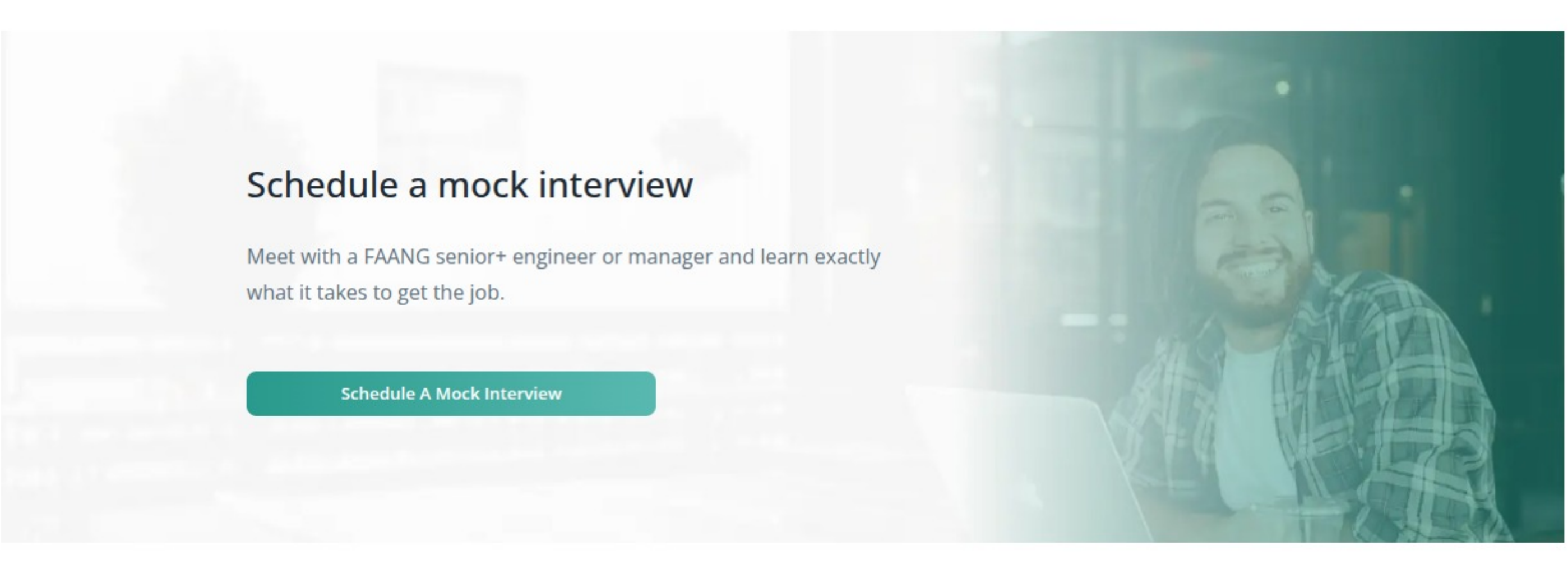

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